Preface

General Approach and Mathematical Level

Our emphasis in creating the ninth edition is less on adding new material and more on providing clarity and deeper understanding. This objective was accomplished in part by including new end-of-chapter material that adds connective tissue between chapters. We affectionately call these comments at the end of the chapter "Pot Holes." They are very useful to remind students of the big picture and how each chapter fits into that picture, and they aid the student in learning about limitations and pitfalls that may result if procedures are misused. A deeper understanding of real-world use of statistics is made available through class projects, which were added in several chapters. These projects provide the opportunity for students alone, or in groups, to gather their own experimental data and draw inferences. In some cases, the work involves a problem whose solution will illustrate the meaning of a concept or provide an empirical understanding of an important statistical result. Some existing examples were expanded and new ones were introduced to create "case studies," in which commentary is provided to give the student a clear understanding of a statistical concept in the context of a practical situation.

In this edition, we continue to emphasize a balance between theory and applications. Calculus and other types of mathematical support (e.g., linear algebra) are used at about the same level as in previous editions. The coverage of analytical tools in statistics is enhanced with the use of calculus when discussion centers on rules and concepts in probability. Probability distributions and statistical inference are highlighted in Chapters 2 through 10. Linear algebra and matrices are very lightly applied in Chapters 11 through 15, where linear regression and analysis of variance are covered. Students using this text should have had the equivalent of one semester of differential and integral calculus. Linear algebra is helpful but not necessary so long as the section in Chapter 12 on multiple linear regression using matrix algebra is not covered by the instructor. As in previous editions, a large number of exercises that deal with real-life scientific and engineering applications are available to challenge the student. The many data sets associated with the exercises are available for download from the website http://www.pearsonhighered.com/datasets.

Summary of the Changes in the Ninth Edition

- Class projects were added in several chapters to provide a deeper understanding of the real-world use of statistics. Students are asked to produce or gather their own experimental data and draw inferences from these data.
- More case studies were added and others expanded to help students understand the statistical methods being presented in the context of a real-life situation. For example, the interpretation of confidence limits, prediction limits, and tolerance limits is given using a real-life situation.
- "Pot Holes" were added at the end of some chapters and expanded in others. These comments are intended to present each chapter in the context of the big picture and discuss how the chapters relate to one another. They also provide cautions about the possible misuse of statistical techniques presented in the chapter.
- Chapter 1 has been enhanced to include more on single-number statistics as well as graphical techniques. New fundamental material on sampling and experimental design is presented.
- Examples added to Chapter 8 on sampling distributions are intended to motivate P-values and hypothesis testing. This prepares the student for the more challenging material on these topics that will be presented in Chapter 10.
- Chapter 12 contains additional development regarding the effect of a single regression variable in a model in which collinearity with other variables is severe.
- Chapter 15 now introduces material on the important topic of response surface methodology (RSM). The use of noise variables in RSM allows the illustration of mean and variance (dual response surface) modeling.
- The central composite design (CCD) is introduced in Chapter 15.
- More examples are given in Chapter 18, and the discussion of using Bayesian methods for statistical decision making has been enhanced.

Content and Course Planning

This text is designed for either a one- or a two-semester course. A reasonable plan for a one-semester course might include Chapters 1 through 10. This would result in a curriculum that concluded with the fundamentals of both estimation and hypothesis testing. Instructors who desire that students be exposed to simple linear regression may wish to include a portion of Chapter 11. For instructors who desire to have analysis of variance included rather than regression, the one-semester course may include Chapter 13 rather than Chapters 11 and 12. Chapter 13 features one-factor analysis of variance. Another option is to eliminate portions of Chapters 5 and/or 6 as well as Chapter 7. With this option, one or more of the discrete or continuous distributions in Chapters 5 and 6 may be eliminated. These distributions include the negative binomial, geometric, gamma, Weibull, beta, and log normal distributions. Other features that one might consider removing from a one-semester curriculum include maximum likelihood estimation,

prediction, and/or tolerance limits in Chapter 9. A one-semester curriculum has built-in flexibility, depending on the relative interest of the instructor in regression, analysis of variance, experimental design, and response surface methods (Chapter 15). There are several discrete and continuous distributions (Chapters 5 and 6) that have applications in a variety of engineering and scientific areas.

Chapters 11 through 18 contain substantial material that can be added for the second semester of a two-semester course. The material on simple and multiple linear regression is in Chapters 11 and 12, respectively. Chapter 12 alone offers a substantial amount of flexibility. Multiple linear regression includes such "special topics" as categorical or indicator variables, sequential methods of model selection such as stepwise regression, the study of residuals for the detection of violations of assumptions, cross validation and the use of the PRESS statistic as well as C_p, and logistic regression. The use of orthogonal regressors, a precursor to the experimental design in Chapter 15, is highlighted. Chapters 13 and 14 offer a relatively large amount of material on analysis of variance (ANOVA) with fixed, random, and mixed models. Chapter 15 highlights the application of two-level designs in the context of full and fractional factorial experiments (2^k). Special screening designs are illustrated. Chapter 15 also features a new section on response surface methodology (RSM) to illustrate the use of experimental design for finding optimal process conditions. The fitting of a second order model through the use of a central composite design is discussed. RSM is expanded to cover the analysis of robust parameter design type problems. Noise variables are used to accommodate dual response surface models. Chapters 16, 17, and 18 contain a moderate amount of material on nonparametric statistics, quality control, and Bayesian inference.

Chapter 1 is an overview of statistical inference presented on a mathematically simple level. It has been expanded from the eighth edition to more thoroughly cover single-number statistics and graphical techniques. It is designed to give students a preliminary presentation of elementary concepts that will allow them to understand more involved details that follow. Elementary concepts in sampling, data collection, and experimental design are presented, and rudimentary aspects of graphical tools are introduced, as well as a sense of what is garnered from a data set. Stem-and-leaf plots and box-and-whisker plots have been added. Graphs are better organized and labeled. The discussion of uncertainty and variation in a system is thorough and well illustrated. There are examples of how to sort out the important characteristics of a scientific process or system, and these ideas are illustrated in practical settings such as manufacturing processes, biomedical studies, and studies of biological and other scientific systems. A contrast is made between the use of discrete and continuous data. Emphasis is placed on the use of models and the information concerning statistical models that can be obtained from graphical tools.

Chapters 2, 3, and 4 deal with basic probability as well as discrete and continuous random variables. Chapters 5 and 6 focus on specific discrete and continuous distributions as well as relationships among them. These chapters also highlight examples of applications of the distributions in real-life scientific and engineering studies. Examples, case studies, and a large number of exercises edify the student concerning the use of these distributions. Projects bring the practical use of these distributions to life through group work. Chapter 7 is the most theoretical chapter

in the text. It deals with transformation of random variables and will likely not be used unless the instructor wishes to teach a relatively theoretical course. Chapter 8 contains graphical material, expanding on the more elementary set of graphical tools presented and illustrated in Chapter 1. Probability plotting is discussed and illustrated with examples. The very important concept of sampling distributions is presented thoroughly, and illustrations are given that involve the central limit theorem and the distribution of a sample variance under normal, independent (i.i.d.) sampling. The t and F distributions are introduced to motivate their use in chapters to follow. New material in Chapter 8 helps the student to visualize the importance of hypothesis testing, motivating the concept of a P-value.

Chapter 9 contains material on one- and two-sample point and interval estimation. A thorough discussion with examples points out the contrast between the different types of intervals—confidence intervals, prediction intervals, and tolerance intervals. A case study illustrates the three types of statistical intervals in the context of a manufacturing situation. This case study highlights the differences among the intervals, their sources, and the assumptions made in their development, as well as what type of scientific study or question requires the use of each one. A new approximation method has been added for the inference concerning a proportion. Chapter 10 begins with a basic presentation on the pragmatic meaning of hypothesis testing, with emphasis on such fundamental concepts as null and alternative hypotheses, the role of probability and the P-value, and the power of a test. Following this, illustrations are given of tests concerning one and two samples under standard conditions. The two-sample t-test with paired observations is also described. A case study helps the student to develop a clear picture of what interaction among factors really means as well as the dangers that can arise when interaction between treatments and experimental units exists. At the end of Chapter 10 is a very important section that relates Chapters 9 and 10 (estimation and hypothesis testing) to Chapters 11 through 16, where statistical modeling is prominent. It is important that the student be aware of the strong connection.

Chapters 11 and 12 contain material on simple and multiple linear regression, respectively. Considerably more attention is given in this edition to the effect that collinearity among the regression variables plays. A situation is presented that shows how the role of a single regression variable can depend in large part on what regressors are in the model with it. The sequential model selection procedures (forward, backward, stepwise, etc.) are then revisited in regard to this concept, and the rationale for using certain P-values with these procedures is provided. Chapter 12 offers material on nonlinear modeling with a special presentation of logistic regression, which has applications in engineering and the biological sciences. The material on multiple regression is quite extensive and thus provides considerable flexibility for the instructor, as indicated earlier. At the end of Chapter 12 is commentary relating that chapter to Chapters 14 and 15. Several features were added that provide a better understanding of the material in general. For example, the end-of-chapter material deals with cautions and difficulties one might encounter. It is pointed out that there are types of responses that occur naturally in practice (e.g. proportion responses, count responses, and several others) with which standard least squares regression should not be used because standard assumptions do not hold and violation of assumptions may induce serious errors. The suggestion is

made that data transformation on the response may alleviate the problem in some cases. Flexibility is again available in Chapters 13 and 14, on the topic of analysis of variance. Chapter 13 covers one-factor ANOVA in the context of a completely randomized design. Complementary topics include tests on variances and multiple comparisons. Comparisons of treatments in blocks are highlighted, along with the topic of randomized complete blocks. Graphical methods are extended to ANOVA to aid the student in supplementing the formal inference with a pictorial type of inference that can aid scientists and engineers in presenting material. A new project is given in which students incorporate the appropriate randomization into each plan and use graphical techniques and P-values in reporting the results. Chapter 14 extends the material in Chapter 13 to accommodate two or more factors that are in a factorial structure. The ANOVA presentation in Chapter 14 includes work in both random and fixed effects models. Chapter 15 offers material associated with 2^k factorial designs; examples and case studies present the use of screening designs and special higher fractions of the 2^k. Two new and special features are the presentations of response surface methodology (RSM) and robust parameter design. These topics are linked in a case study that describes and illustrates a dual response surface design and analysis featuring the use of process mean and variance response surfaces.

Computer Software

Case studies, beginning in Chapter 8, feature computer printout and graphical material generated using both SAS and MINITAB. The inclusion of the computer reflects our belief that students should have the experience of reading and interpreting computer printout and graphics, even if the software in the text is not that which is used by the instructor. Exposure to more than one type of software can broaden the experience base for the student. There is no reason to believe that the software used in the course will be that which the student will be called upon to use in practice following graduation. Examples and case studies in the text are supplemented, where appropriate, by various types of residual plots, quantile plots, normal probability plots, and other plots. Such plots are particularly prevalent in Chapters 11 through 15.

Supplements

Instructor's Solutions Manual. This resource contains worked-out solutions to all text exercises and is available for download from Pearson Education's Instructor Resource Center.

Student Solutions Manual ISBN-10: 0-321-64013-6; ISBN-13: 978-0-321-64013-0. Featuring complete solutions to selected exercises, this is a great tool for students as they study and work through the problem material.

PowerPoint® Lecture Slides ISBN-10: 0-321-73731-8; ISBN-13: 978-0-321-73731-1. These slides include most of the figures and tables from the text. Slides are available to download from Pearson Education's Instructor Resource Center.